

LECTURE 7

Applications of the Bernoulli Equation

Reference: Munson, Rothmayer, Okiishi & Huebsch, Fundamentals of Fluid Mechanics, Chapter 3
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Static, stagnation, dynamic and total pressure

Every term of the Bernoulli equation has the dimension dimension of force per unit area, and each is a form of of pressure:

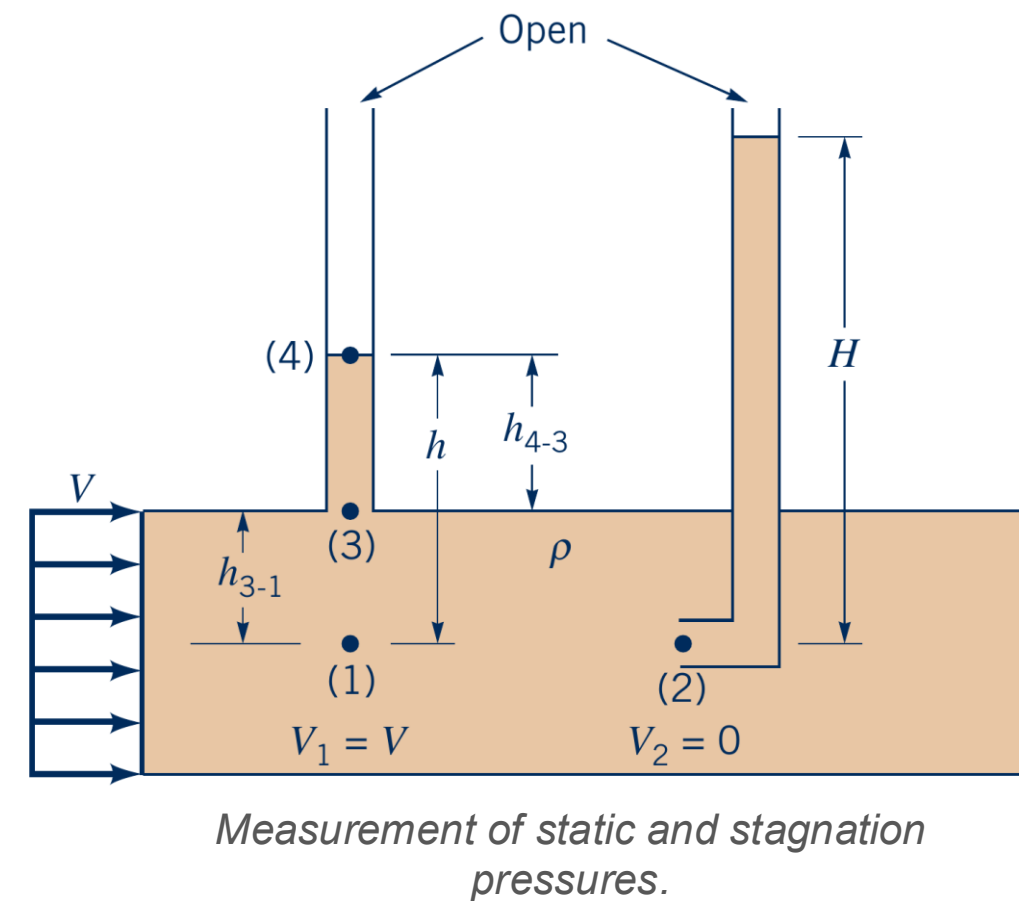
$$\underbrace{p}_{\text{static}} + \underbrace{\frac{1}{2}\rho v^2}_{\text{dynamic}} + \underbrace{\gamma z}_{\text{hydrostatic}} = \underbrace{p_T}_{\text{total}}$$

p is the thermodynamic pressure of the fluid as it flows, flows, the static pressure.

To measure it, move with the fluid — or drill a hole in a a flat surface and fit a piezometer tube, point (3) in the the figure.

γz is the hydrostatic pressure. It is not really a pressure, but the change in pressure that elevation can produce.

Each term in the Bernoulli equation can be interpreted as a form of pressure.



Stagnation points

Put a small tube into the flow pointing upstream

The fluid at its tip, point (2), is brought to rest:

$v_2 = 0$, and (2) is a **stagnation point**. With $z_1 = z_2$,

$$p_2 = p_1 + \frac{1}{2}\rho v_1^2$$

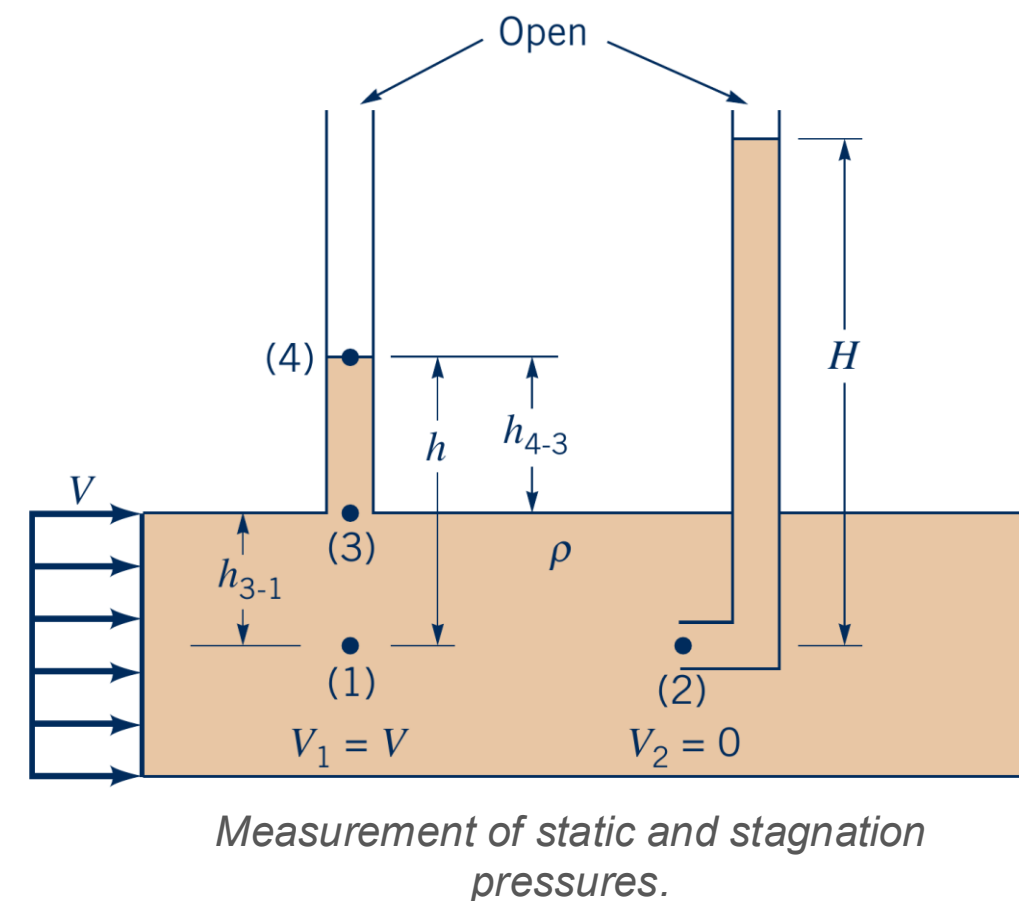
The pressure there exceeds the static pressure by the dynamic pressure $\rho v_1^2/2$

There is a stagnation point on every stationary body placed in a flow; the dividing streamline that ends on it is the stagnation streamline.

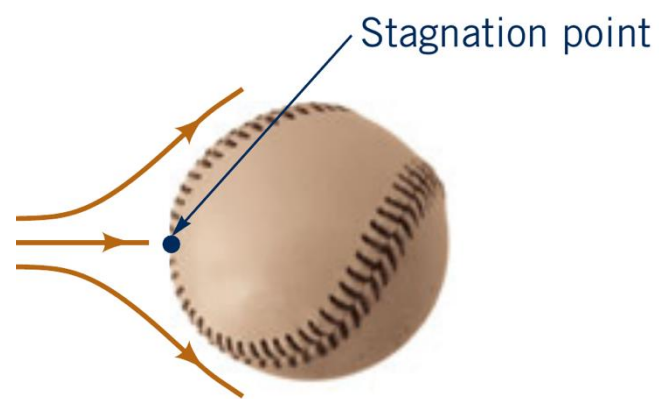
If elevation is neglected, the stagnation pressure is the largest pressure obtainable along a streamline: all of the kinetic energy has become a pressure rise

The sum of the 3 is the total pressure, constant along a streamline:

$$p + \frac{1}{2}\rho v^2 + \gamma z = p_T = \text{constant along a streamline}$$



Stagnation points



(a)



(b)

On a symmetric body like the ball in (a) it sits right at the front.

On the car, it is where the jet strikes.

The **stagnation pressure** is the pressure obtained when a flowing fluid is decelerated to zero speed by a frictionless process.

Stagnation point flow

On any body in a flowing fluid, there is a stagnation point.

Some of the fluid flows over the body and some under it, and the dividing line — the stagnation streamline — ends at that point.



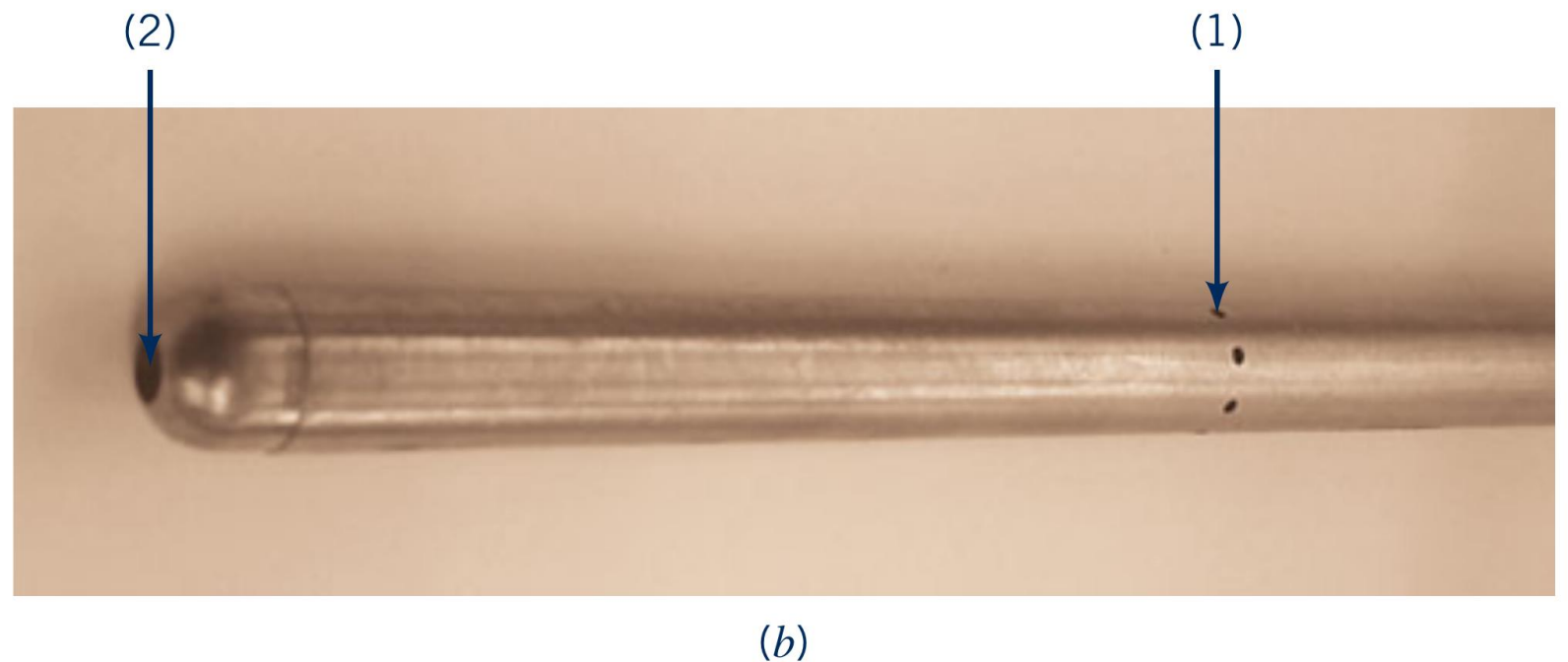
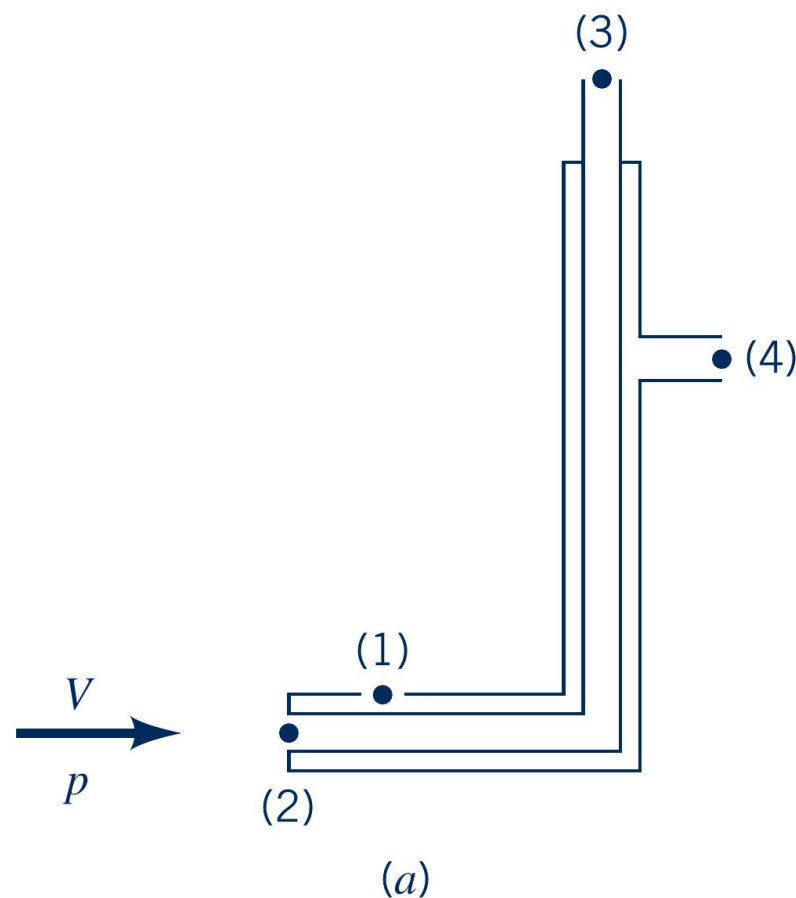
Video © B. R. Munson

The Pitot-static tube

Knowing both the static and the stagnation pressure gives the flow speed.

This is the principle of the **Pitot-static tube**, named after Henri de Pitot (1695–1771). Two concentric tubes feed a differential gage:

The Pitot-static tube.



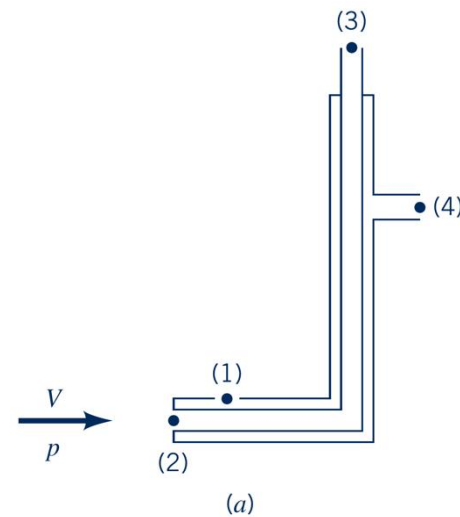
Stagnation pressure:

$$p_3 = p + \frac{1}{2}\rho v^2,$$

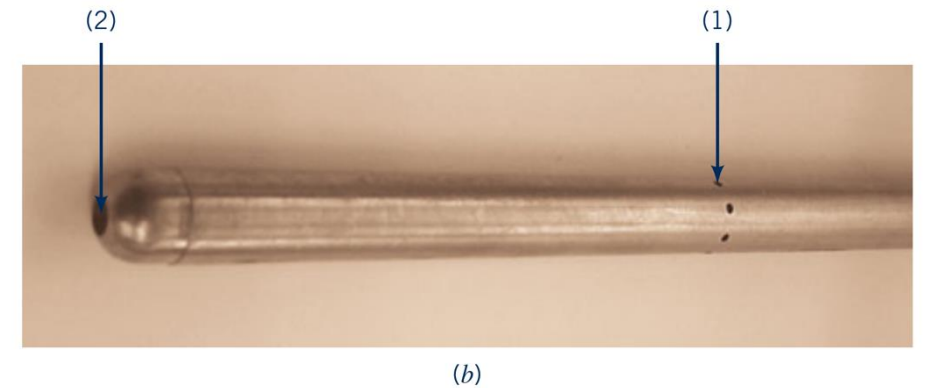
Static pressure:

$$p_4 = p_1 = p$$

The Pitot-static tube



The Pitot-static tube.



The centre tube reads the stagnation pressure at its tip

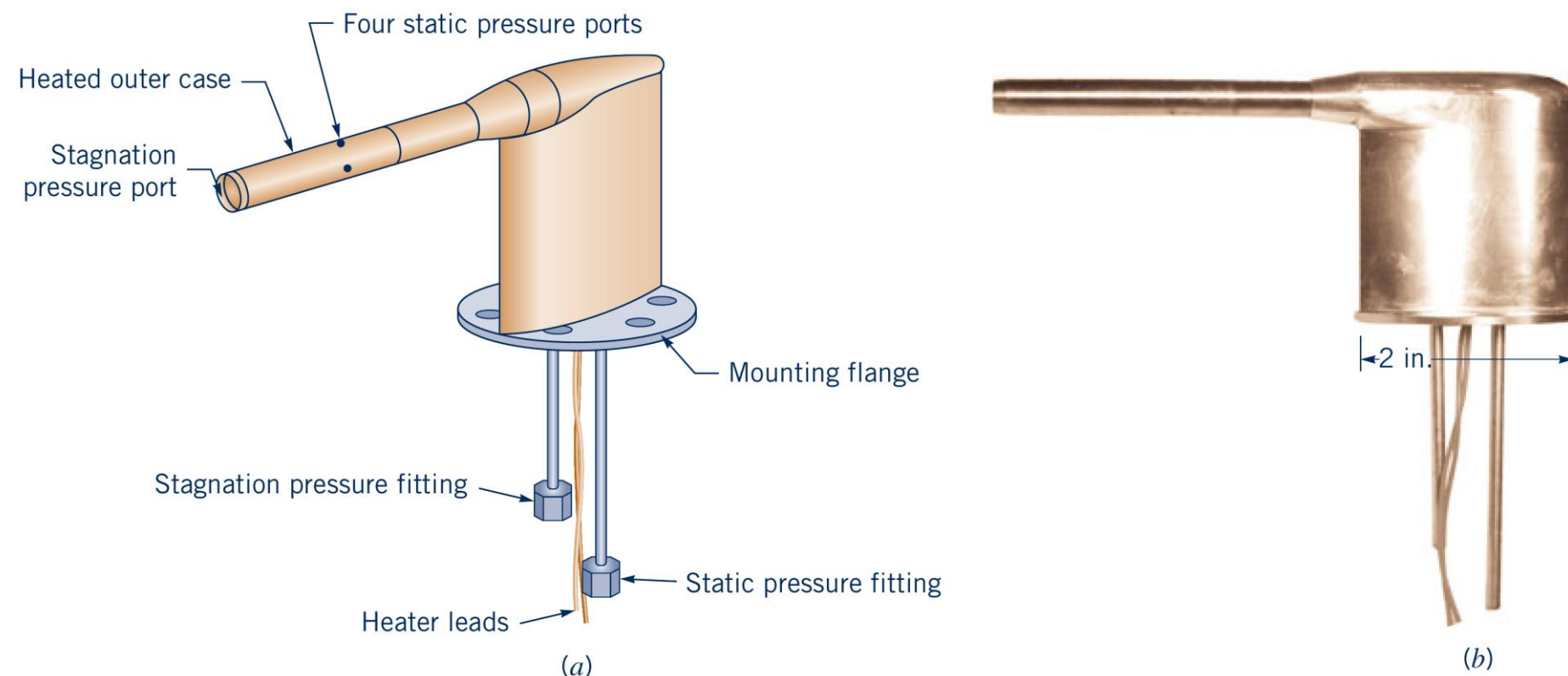
The outer tube has small side holes that read the static pressure. Subtracting,

$$v = \sqrt{\frac{2 (p_3 - p_4)}{\rho}}$$

Pitot-static tubes measure fluid velocity by converting velocity into pressure.

The aircraft probe

The shapes and sizes vary considerably. A probe used to measure aircraft airspeed carries a heated outer case, a stagnation pressure port at the tip and four static pressure ports along the side.



Airplane Pitot-static probe. (a) Schematic, (b) Photograph. (Photograph courtesy of Aero-Instruments Co., LLC.)

Accurate measurement of static pressure requires great care.

The static holes must be smooth and free of burrs: any imperfection converts some of the kinetic energy into a pressure rise and the reading is wrong.

Blocked Pitot tubes — a forgotten cover, or an insect nest — have caused fatal accidents.

The airspeed indicator

The stagnation pressure at the tip of a Pitot tube is greater than the static pressure by one half times the fluid density times the speed squared. Thus, the speed can be determined by measuring the pressure difference.

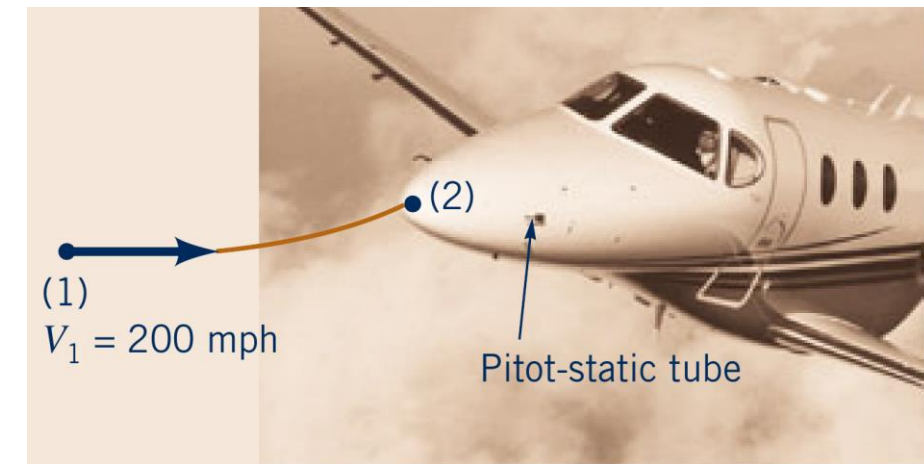


Video © B. R. Munson

The Pitot tube on an aircraft is connected to the airspeed indicator, a pressure transducer calibrated to read in knots or miles per hour rather than in units of pressure. Because the air density varies with temperature, pressure and altitude, the speed displayed — the indicated airspeed — equals the speed at which the aircraft is actually flying through the air only at standard density.

Example: an aircraft Pitot-static tube

An aeroplane flies at 89.4 m/s (200 mph) at an elevation of 3048 m (10 000 ft) in a standard atmosphere. Find the pressure far ahead of the aircraft, the pressure at the stagnation point on its nose, and the difference a Pitot-static probe would read.



(Photo courtesy of Hawker Beechcraft Corporation.)

From the standard atmosphere table, the static pressure far ahead of the aircraft is:

$$z = 3048 \text{ m} : \quad p_1 = 69.7 \text{ kPa (abs)}, \quad \rho = 0.905 \text{ kg/m}^3, \quad v_1 = 89.4 \text{ m/s}$$

With the coordinate system fixed to the aircraft, $v_2 = 0$ and our equation gives

$$\begin{aligned} p_2 &= p_1 + \frac{\rho v_1^2}{2} = 69.7 \times 10^3 + \frac{(0.905)(89.4)^2}{2} \\ &= (69.7 + 3.6) \text{ kPa} = 73.3 \text{ kPa (abs)} \end{aligned}$$

$$p_2 - p_1 = \frac{\rho v_1^2}{2} = 3.6 \text{ kPa}$$

Was the incompressible assumption fair?

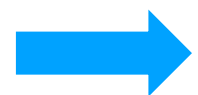
We assumed the density stays constant from (1) to (2).

But $\rho = p/RT$, so a change in pressure changes the density. At this speed, the ratio of the absolute pressures is close to one,

$$\frac{p_1}{p_2} = \frac{69.7}{73.3} = 0.951$$

so the density change is negligible and the answer stands.

Repeating the calculation at higher speeds shows the ratio falling steadily:
at the 800 to 900 km/h flown by commercial airliners the density change matters and compressible flow methods are needed.



Every use of the Bernoulli equation should end with this check!

Using the Bernoulli equation

Between any two points (1) and (2) on a streamline in steady, inviscid, incompressible flow,

$$p_1 + \frac{1}{2}\rho v_1^2 + \gamma z_1 = p_2 + \frac{1}{2}\rho v_2^2 + \gamma z_2$$

Six variables appear. If five are known, the sixth follows.

In many problems that is not enough on its own, and a second equation — **conservation of mass** — is needed alongside it.

Applications:

- **free jets** — the flow of a liquid from a large reservoir
- **confined flows** — nozzles and pipes, where the pressure cannot be prescribed in advance
- **flowrate measurement** — orifice, nozzle and Venturi meters, sluice gates and weirs

Free jets

A jet of liquid of diameter d leaves a nozzle at the bottom of a large reservoir. Apply Bernoulli's eq. between (1) and (2): the reservoir is large so $v_1 \approx 0$, it is open so $p_1 = 0$ gage, the jet is free so $p_2 = 0$, and $z_1 = h$, $z_2 = 0$. Then

$$\gamma h = \frac{1}{2} \rho v^2$$

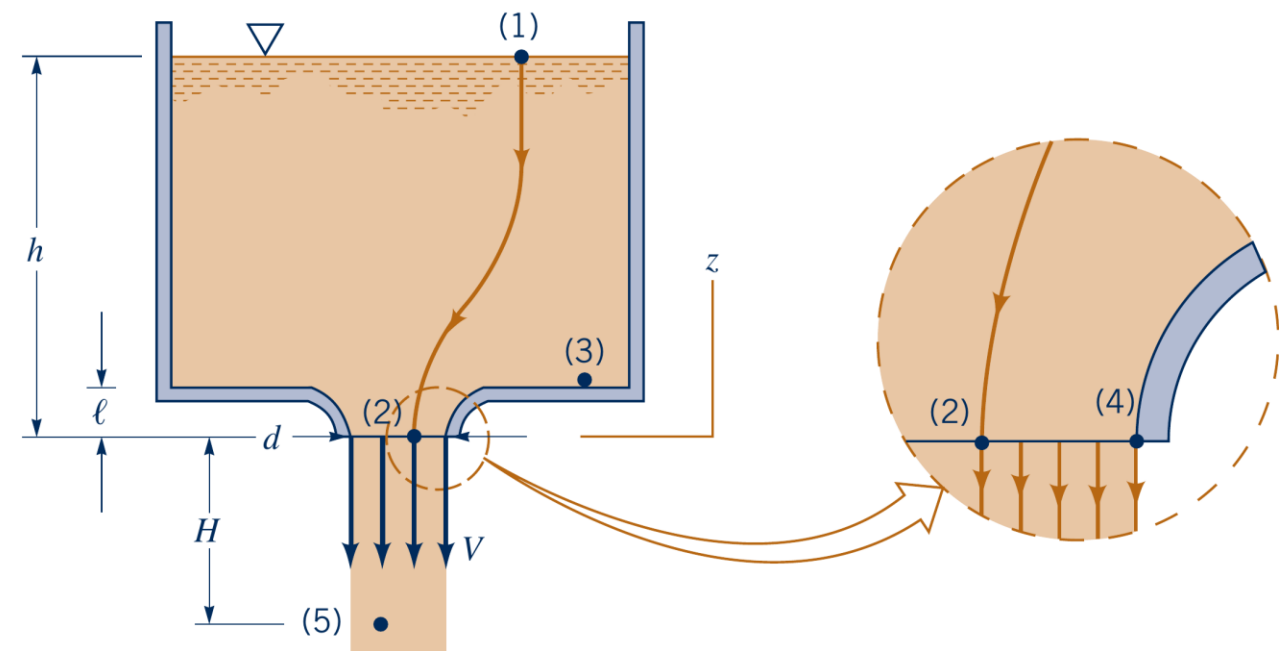
$$v = \sqrt{\frac{2\gamma h}{\rho}} = \sqrt{2gh}$$

the result Torricelli obtained in 1643. It is the speed any object dropped from rest reaches after falling a distance h : the elevation head at (1) has become the velocity head at (2).

Outside the nozzle the jet keeps falling with zero pressure throughout, and after a further distance H the speed is

$$v = \sqrt{2g(h + H)}$$

Vertical flow from a tank.



Flow from a tank

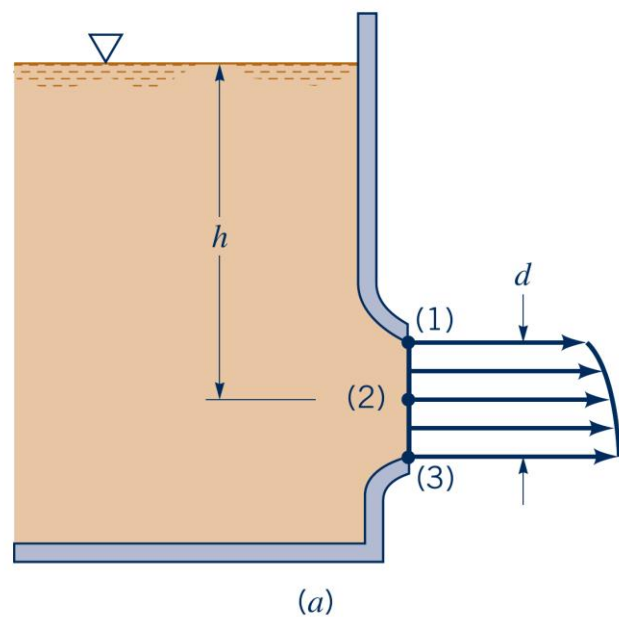
According to the Bernoulli equation, the velocity of a fluid flowing through a hole in the side of an open tank or reservoir is proportional to the square root of the depth of fluid above the hole.



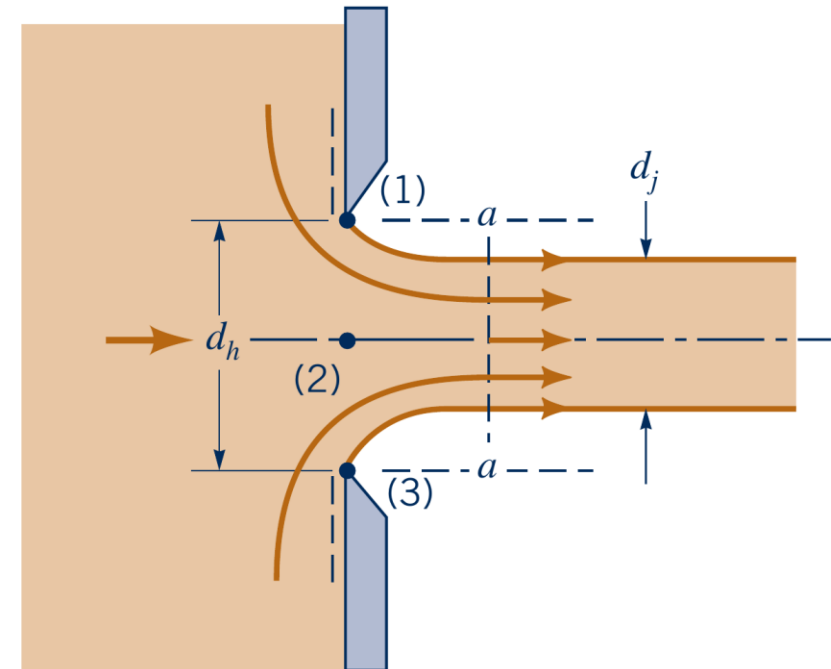
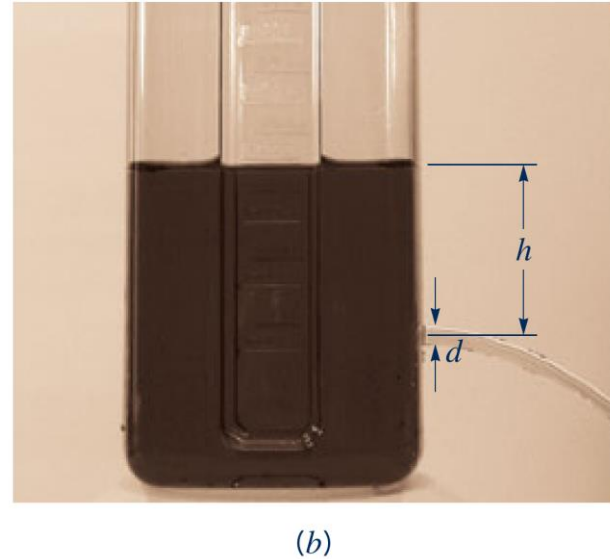
Video courtesy of U.S. Bureau of Reclamation; video still © B. R. Munson

The velocity of the jets leaving an open bottle with four holes in it is clearly related to the depth of water above each hole: **the greater the depth, the higher the velocity.**

The horizontal nozzle and the vena contracta



Horizontal flow from a tank.



Vena contracta effect for a sharp-edged orifice.

For a horizontal nozzle the centreline speed is slightly greater than at the top and slightly less than at the bottom.

In general, $d \ll h$, so the centreline value is a fair average.

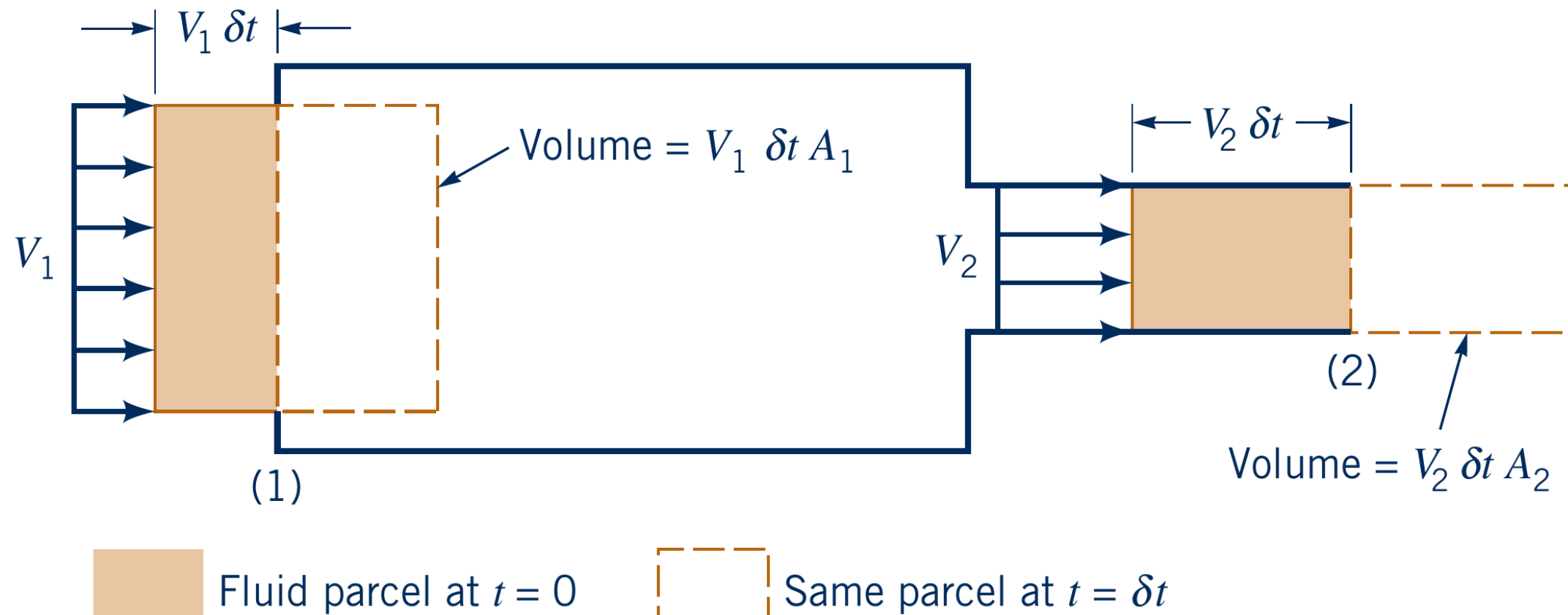
If the exit is a flat plate rather than a contoured nozzle, the fluid cannot turn the sharp 90° corner and the jet diameter is smaller than the hole:

$$C_c = \frac{A_j}{A_h} < 1$$

The ratio of the areas is the contraction coefficient.

The diameter of a fluid jet is often smaller than that of the hole from which it flows.

Confined flows and the continuity equation

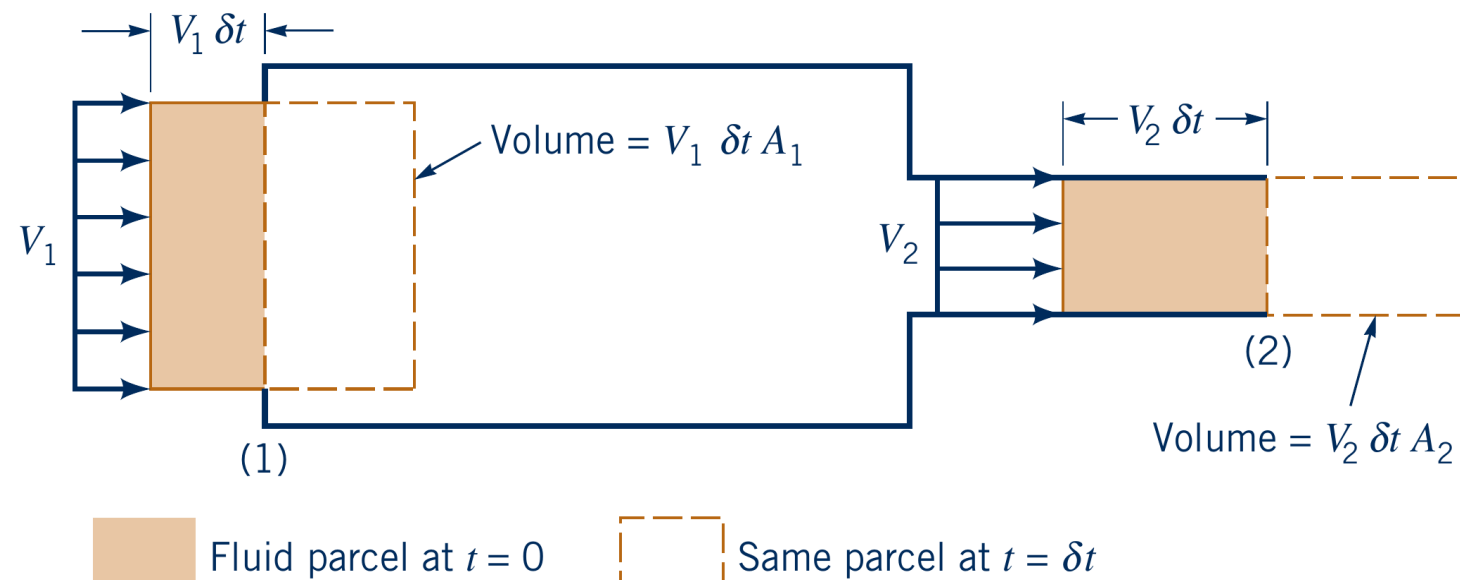


Steady flow into and out of a volume.

When the fluid is confined inside a device its pressure cannot be prescribed in advance, and a second equation is needed.

For steady flow through a fixed volume with one inlet and one outlet, **the rate in must equal the rate out.**

Confined flows and the continuity equation



The **mass flowrate** and **volume flowrate** through an area A are:

$$\dot{m} = \rho Q = \rho v A, \quad Q = v A$$

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2$$

and if the density is constant this becomes the **continuity equation for incompressible flow**:

$$A_1 v_1 = A_2 v_2, \quad \text{or} \quad Q_1 = Q_2$$

The continuity equation states that mass cannot be created or destroyed.

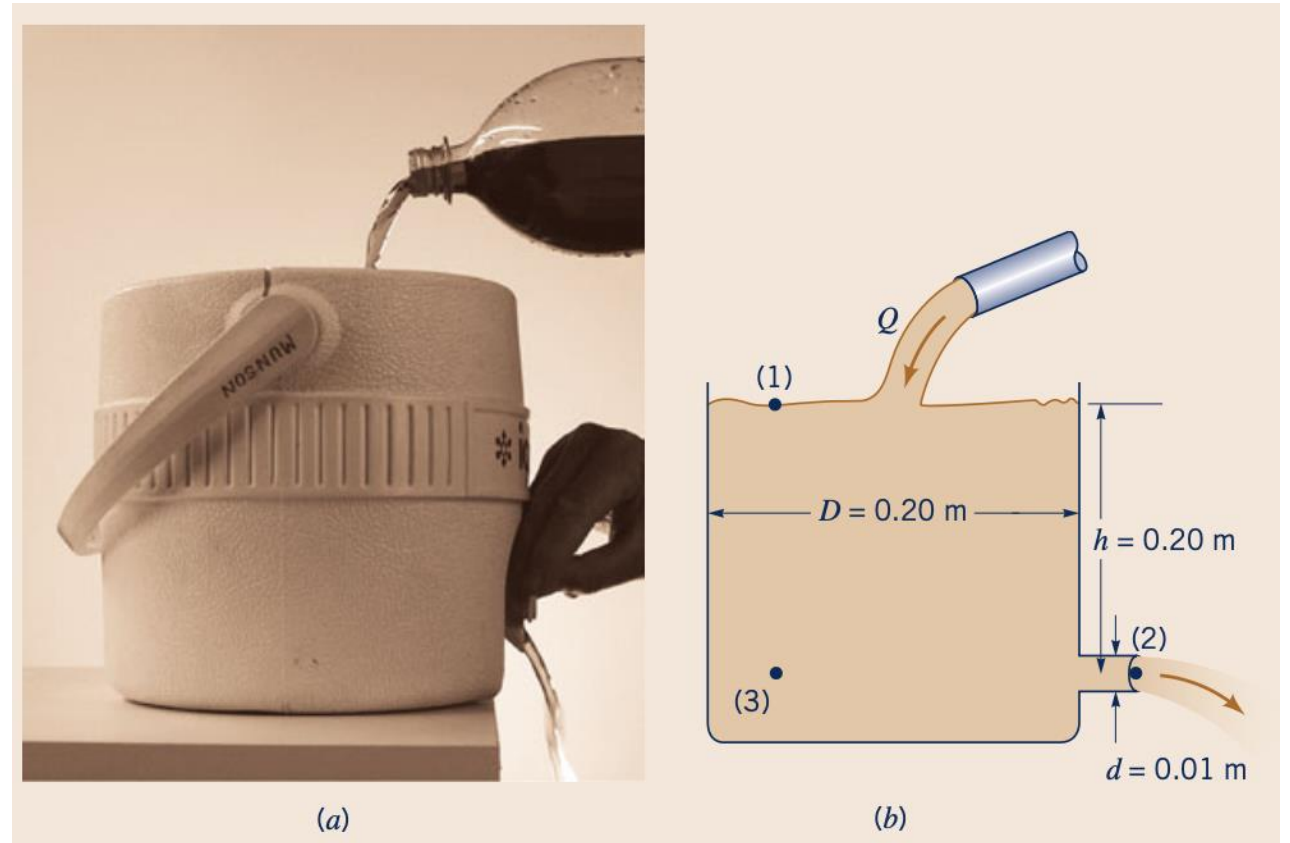
Example: gravity-driven flow from a tank

A stream of beverage of diameter $d = 0.01 \text{ m}$ flows from a cooler of diameter $D = 0.20 \text{ m}$.

Find the flowrate Q that keeps the depth constant at $h = 0.20 \text{ m}$.

With $p_1 = p_2 = 0$, $z_1 = h$ and $z_2 = 0$, Bernoulli's eq. divided by ρ gives

$$\frac{1}{2}v_1^2 + gh = \frac{1}{2}v_2^2$$



The level is constant, but there is still an average velocity v_1 across (1). Continuity gives:

$$\frac{\pi}{4}D^2v_1 = \frac{\pi}{4}d^2v_2 \quad \Rightarrow \quad v_1 = \left(\frac{d}{D}\right)^2 v_2$$

$$v_2 = \sqrt{\frac{2gh}{1 - (d/D)^4}} = \sqrt{\frac{2(9.81)(0.20)}{1 - (0.01/0.20)^4}} = 1.98 \text{ m/s}$$

$$Q = A_2v_2 = \frac{\pi}{4}(0.01)^2(1.98) = 1.56 \times 10^{-4} \text{ m}^3/\text{s}$$

Cavitation

The Bernoulli equation gives one route to cavitation

Reduce the flow area, as in a kinked garden hose, and the velocity rises and the pressure falls.

If it falls to the vapour pressure of the liquid, the liquid boils — at ordinary temperature.



Cavitation occurs when the pressure is reduced to the vapour pressure.

Vapour bubbles form, are carried into a region of higher pressure and collapse. The implosions produce very large local pressure transients — believed to reach ~ 700 MPa — and they occur against the nearest solid surface.

Cavitation can cause damage to equipment.

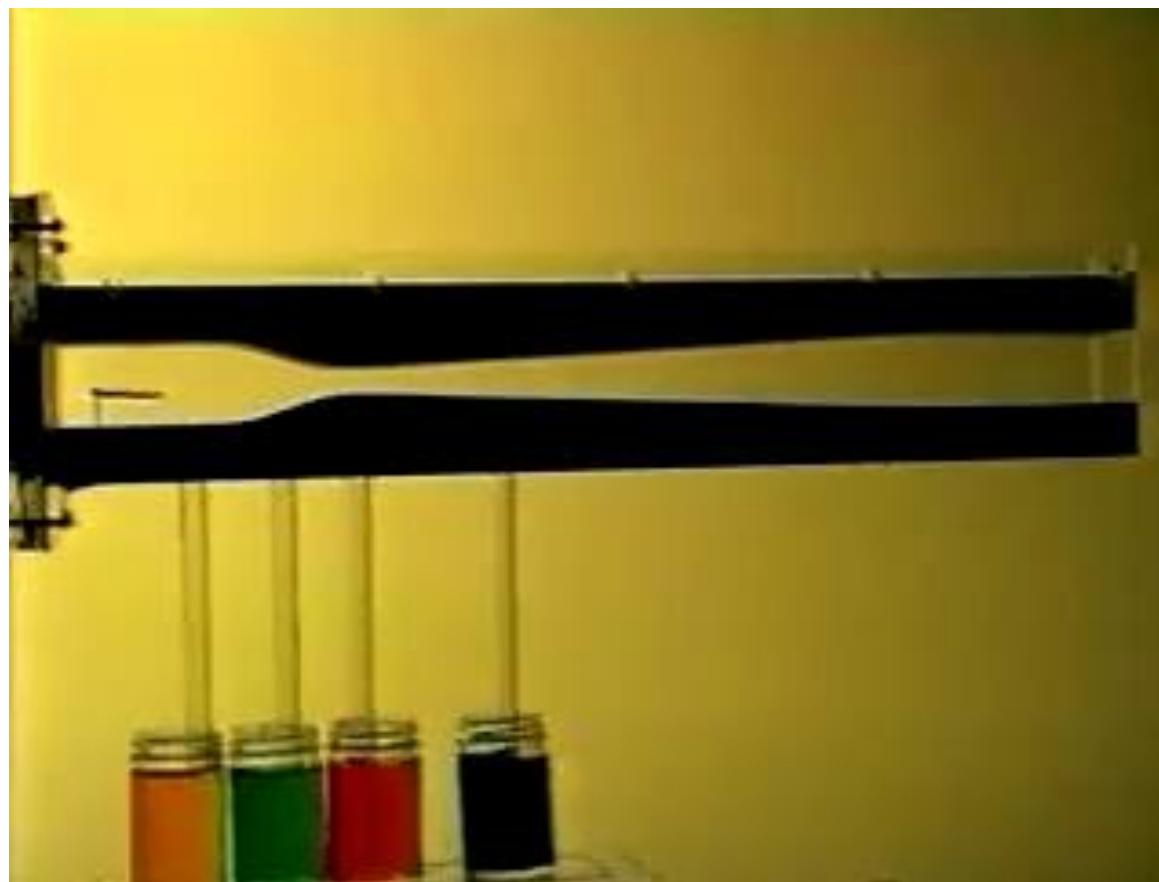
The hissing of a kinked hose is cavitation you can hear. The pitting of ship propellers and pump impellers is cavitation you can sense.

The Venturi channel

As a fluid flows through a converging channel, the pressure is reduced in accordance with the continuity and the Bernoulli equations.

As continuity predicts, the velocity of the air increases because the channel area is reduced.

As Bernoulli predicts, that increase in velocity causes a decrease in pressure. The attached water columns show that the greatest pressure reduction occurs at the narrowest part of the channel.



Video © B. R. Munson

The same principle is used in a garden sprayer, so that liquid chemicals can be sucked out of the bottle and mixed with the water in the hose.

Flowrate measurement

Place a restriction in the pipe and measure the pressure difference between:

- the slow, high-pressure section (1) and
- the fast, low-pressure section (2).

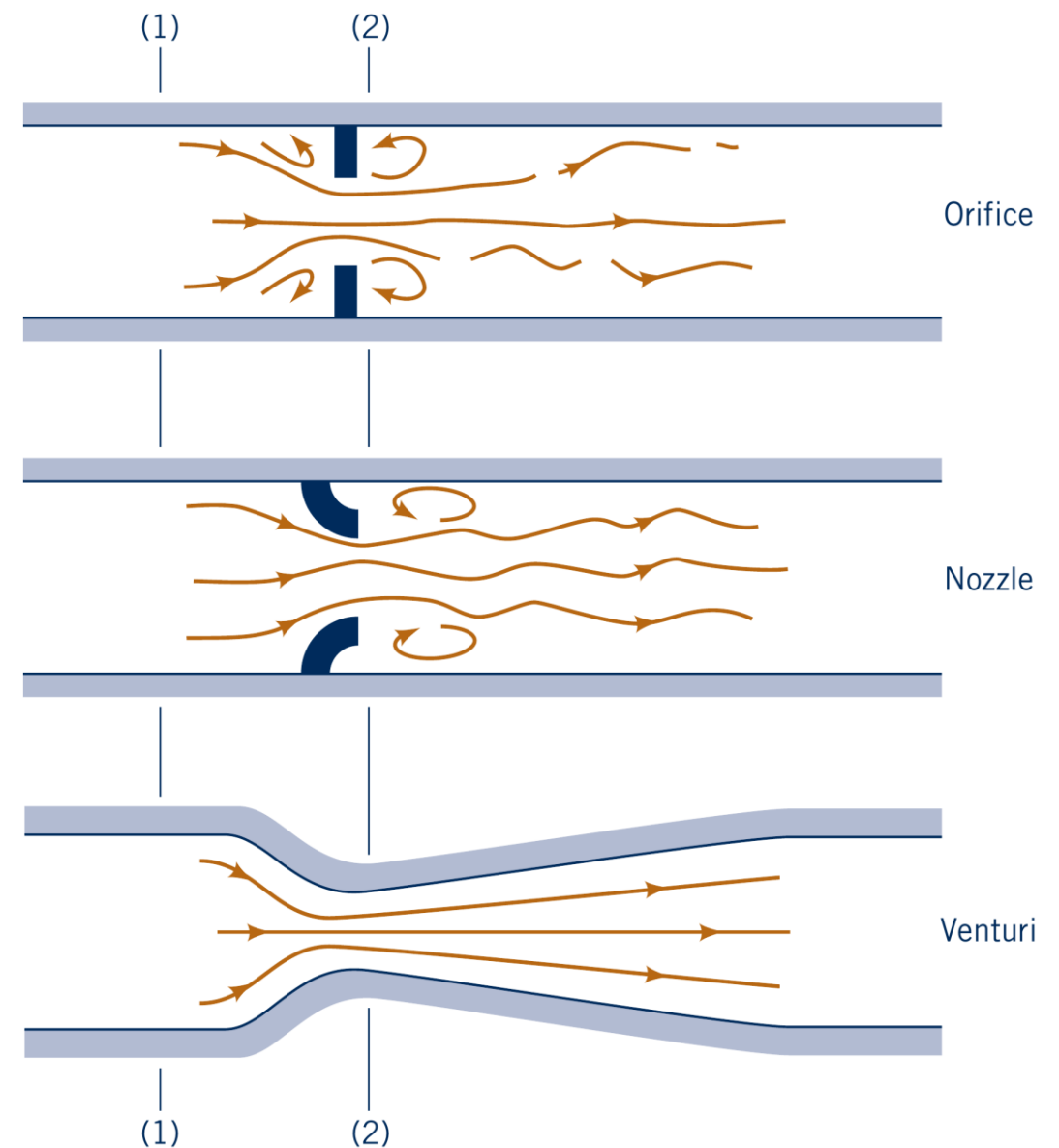
The **orifice**, the **nozzle** and the **Venturi** all work on this principle: they differ in cost, in accuracy, and in how closely they obey the ideal assumptions.

Take the flow horizontal, steady, inviscid and incompressible.

Combining the Bernoulli equation with $Q = A_1 v_1 = A_2 v_2$ gives the **theoretical flowrate**

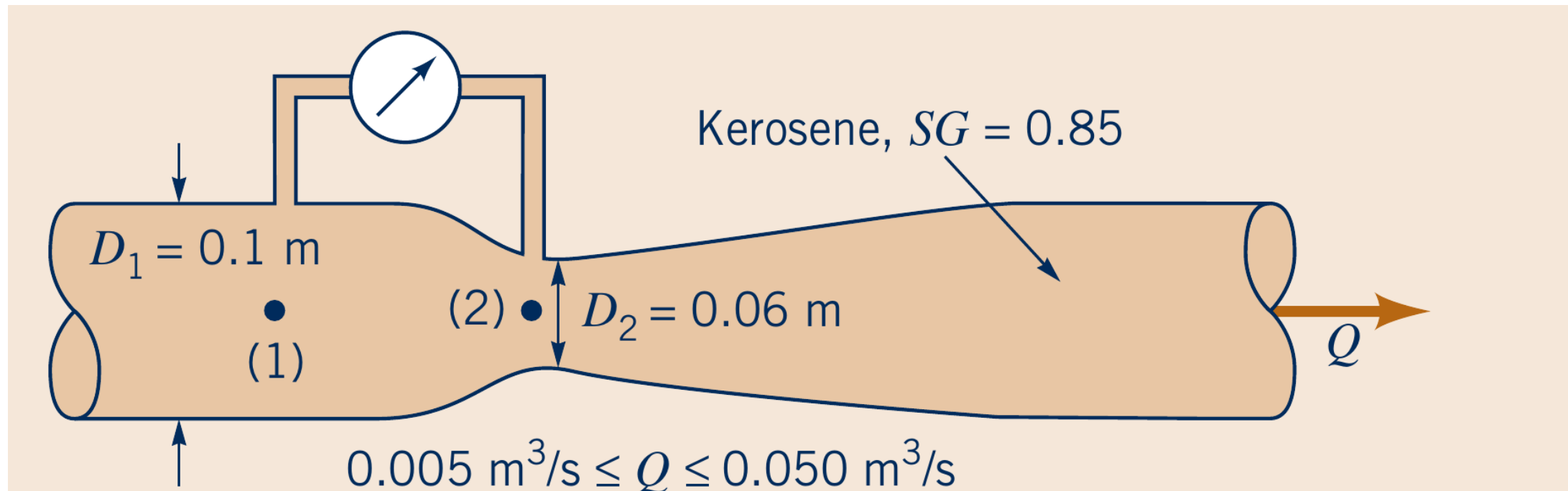
$$Q = A_2 \sqrt{\frac{2(p_1 - p_2)}{\rho [1 - (A_2/A_1)^2]}}$$

Typical devices for measuring flowrate in pipes.



The flowrate varies as the square root of the pressure difference across the flowmeter.

Example: a Venturi meter



Kerosene of $SG = 0.85$ flows through the meter at between 0.005 and $0.050 \text{ m}^3/\text{s}$. Find the range of $p_1 - p_2$ needed to measure it.

$$\frac{A_2}{A_1} = \left(\frac{0.06}{0.10} \right)^2 = 0.36$$

and for the smallest flowrate

$$p_1 - p_2 = (0.005)^2 (850) \frac{1 - 0.36^2}{2 [(\pi/4)(0.06)^2]^2} = 1160 \text{ N/m}^2 = 1.16 \text{ kPa}$$

$$p_1 - p_2 = \frac{Q^2 \rho [1 - (A_2/A_1)^2]}{2A_2^2}$$

$$\rho = SG \rho_{\text{H}_2\text{O}} = 0.85(1000) = 850 \text{ kg/m}^3,$$

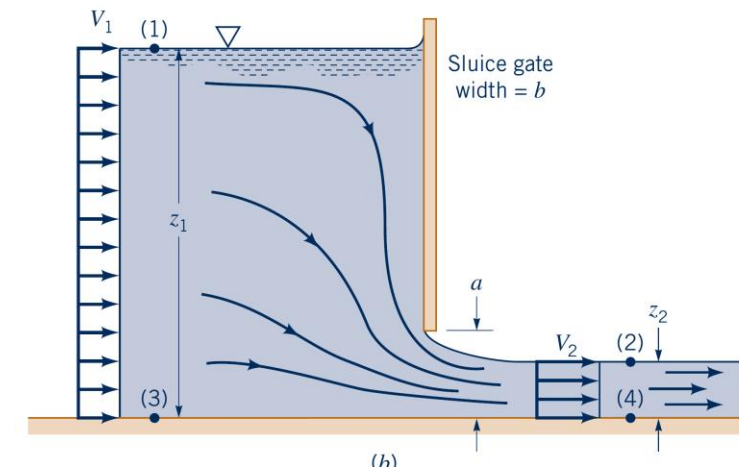
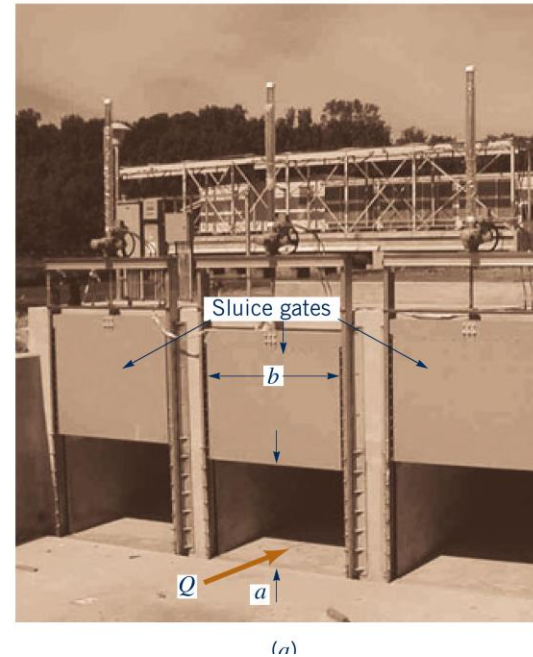
$$1.16 \text{ kPa} \leq p_1 - p_2 \leq 116 \text{ kPa}$$

A ten-fold increase in flowrate needs a hundred-fold increase in pressure difference.

Sluice gates

Sluice gates regulate and measure the flowrate in open channels. Apply the Bernoulli equation between the free surfaces at (1) and (2), where $p_1 = p_2 = 0$, with $Q = bv_1z_1 = bv_2z_2$:

$$Q = z_2 b \sqrt{\frac{2g(z_1 - z_2)}{1 - (z_2/z_1)^2}}$$



Sluice gate geometry. (Photograph courtesy of Plasti-Fab, Inc.)

and if $z_1 \gg z_2$ the upstream kinetic energy is negligible and

$$z_1 \gg z_2 : \quad Q = z_2 b \sqrt{2gz_1}$$

It is the downstream depth z_2 that enters, not the gate opening a : the fluid cannot turn the sharp corner, so a **vena contracta** forms.

$$C_c = \frac{z_2}{a} \simeq 0.61 \quad (0 < a/z_1 < 0.2)$$

The flowrate under a sluice gate depends on the water depths on either side of the gate.

A sluice gate

Water flows under the sluice gate shown.

Find the approximate flowrate per unit width of the channel.

Here $z_1 = 5.0 \text{ m}$ and $a = 0.80 \text{ m}$, so

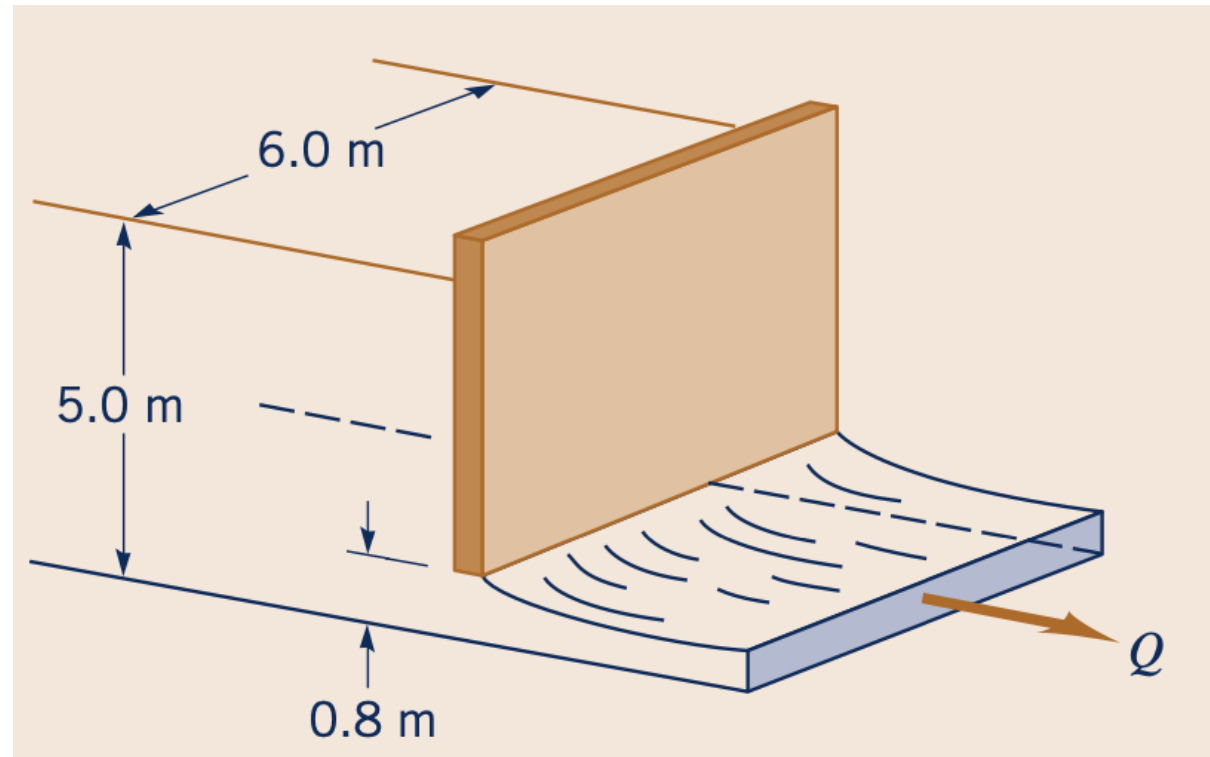
$$\frac{a}{z_1} = \frac{0.80}{5.0} = 0.16 < 0.20 \quad \Rightarrow \quad z_2 = C_c a = 0.61(0.80) = 0.488 \text{ m}$$

$$\frac{Q}{b} = (0.488) \sqrt{\frac{2(9.81)(5.0 - 0.488)}{1 - (0.488/5.0)^2}} = 4.61 \text{ m}^2/\text{s}$$

Neglecting the upstream kinetic energy instead would have given

$$\frac{Q}{b} = z_2 \sqrt{2gz_1} = 0.488 \sqrt{2(9.81)(5.0)} = 4.83 \text{ m}^2/\text{s}$$

The difference is small because the depth ratio is large, $z_1/z_2 = 10.2$. Note that the flowrate is not proportional to the depth: doubling z_1 from 5 m to 10 m raises Q/b only from 4.61 to 6.67 m^2/s .



Restrictions on the Bernoulli equation

Compressibility

The density must be known as a function of the pressure before the first term can be integrated.

$$\int \frac{dp}{\rho} + \frac{1}{2}v^2 + gz = \text{constant}$$

Rotational effects

The constant is the same on every streamline only when the flow is irrotational:

$$\nabla \times \mathbf{v} = 0$$

Unsteady effects

A term in the local acceleration survives along the streamline.

Other restrictions

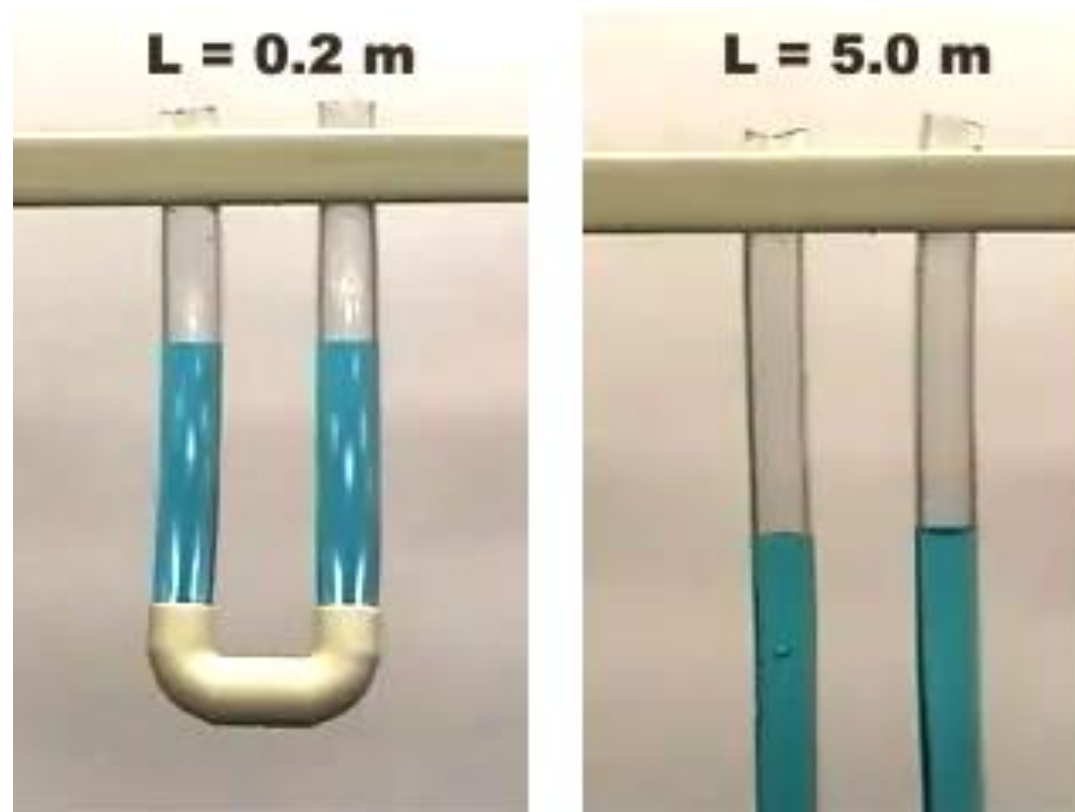
No viscous dissipation, and no pump, turbine or fan anywhere between the two points.

Unsteady effects: oscillations in a U-tube

Many flows are steady. Others are not precisely steady, but their unsteadiness is of secondary importance and they can be treated as quasi-steady. For a third group the unsteadiness is the whole behaviour.

$$p_1 + \frac{1}{2}\rho v_1^2 + \gamma z_1 + \rho \int_{s_1}^{s_2} \frac{\partial v}{\partial t} ds = p_2 + \frac{1}{2}\rho v_2^2 + \gamma z_2$$

The oscillation of a liquid in a U-tube is one of the third kind. Released from a state with one end higher than the other, the column swings back and forth much as a simple pendulum does. Because of viscous effects inside the tube the oscillations finally die out, just as a pendulum stops swinging because of the drag on it.

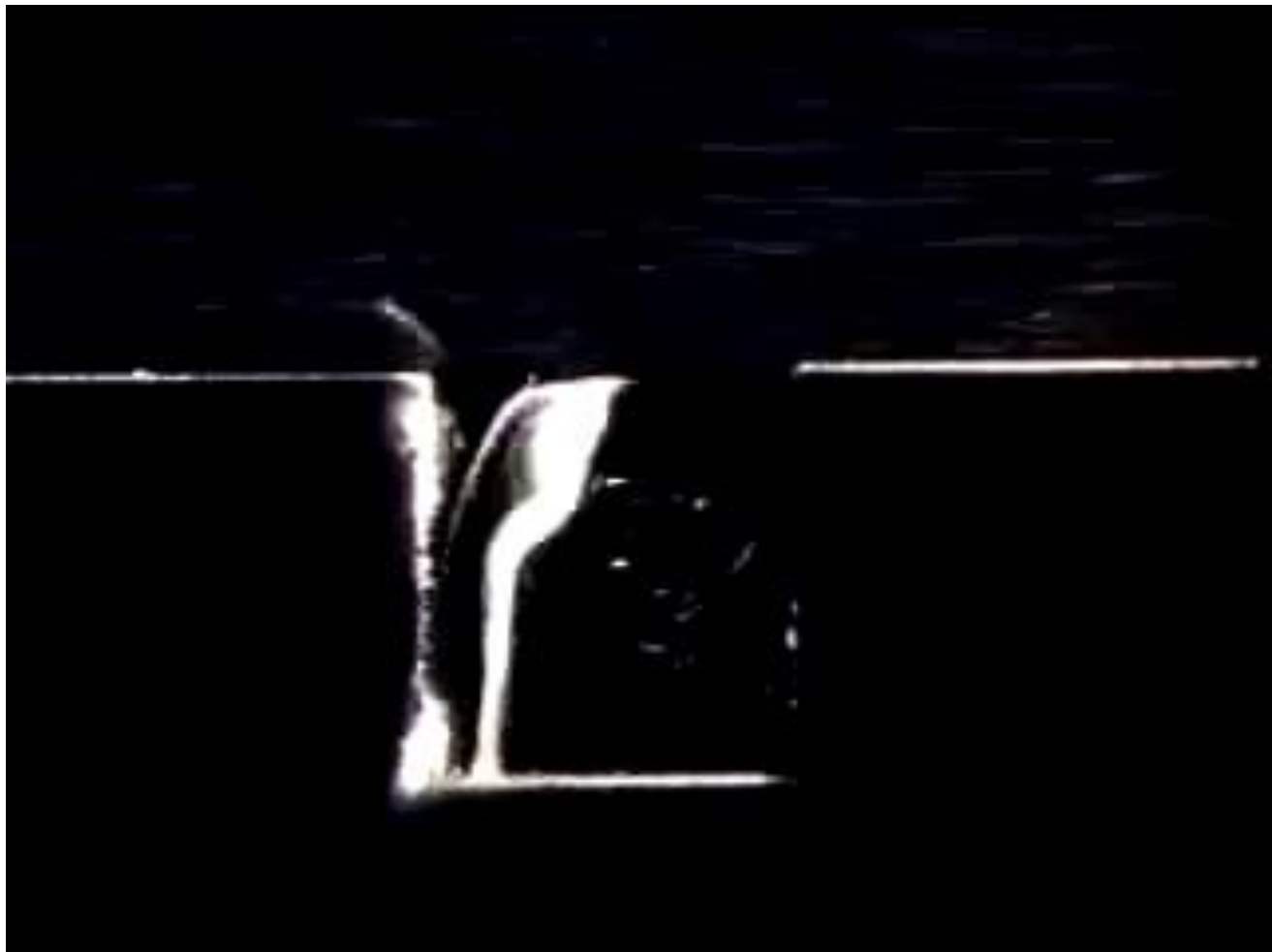


Video © B. R. Munson

Rotational effects: flow over a cavity

In some situations, it is valid to apply the Bernoulli equation across streamlines. If the flow is rotational, trying to doing so will give the wrong answer.

$$p + \frac{1}{2}\rho v^2 + \gamma z = C_1 \neq C_2$$



The swirling flow in the cavity shown is produced by the horizontal stream running across its open top.

The viscous shear stress at the interface drags the cavity fluid along and sets it circulating.

On top of that overall rotation, each fluid particle at the interface is itself rotating — so the Bernoulli equation may not be carried across the streamlines from the main flow into the cavity.

Lecture summary

We have studied the Bernoulli equation and its applications.

$$p + \frac{1}{2}\rho v^2 = p_{\text{stag}}$$

static + dynamic =
stagnation

$$v = \sqrt{\frac{2(p_3 - p_4)}{\rho}}$$

the Pitot-static tube

$$v = \sqrt{2gh}$$

a free jet

$$Q = A_1 v_1 = A_2 v_2$$

continuity

$$Q = A_2 \sqrt{\frac{2(p_1 - p_2)}{\rho(1 - \beta^4)}}$$

orifice, nozzle, Venturi

$$Q \simeq z_2 b \sqrt{2gz_1}$$

a sluice gate

$$\frac{1}{2}\rho v_2^2 (1 - \beta^4) \leq p_1 - p_v$$

how fast before the throat
cavitates

$$\nabla \times \mathbf{v} = 0$$

a constant for every
streamline